LAPORAN MODUL 12
PRAKTIKUM PERMODELAN STATISTIKA

Disusun oleh :

Nama : IRVAN CAHYA NUGRAHA

NIM : 245410034

Kelas : IF-1

PROGRAM STUDI INFORMATIKA

PROGRAM SARJANA

FAKULTAS TEKNOLOGI INFORMASI

UNIVERSITAS TEKNOLOGI DIGITAL INDONESIA

YOGYAKARTA

2024/2025

LAPORAN PRAKTIKUM
MODUL 12
ARIMA

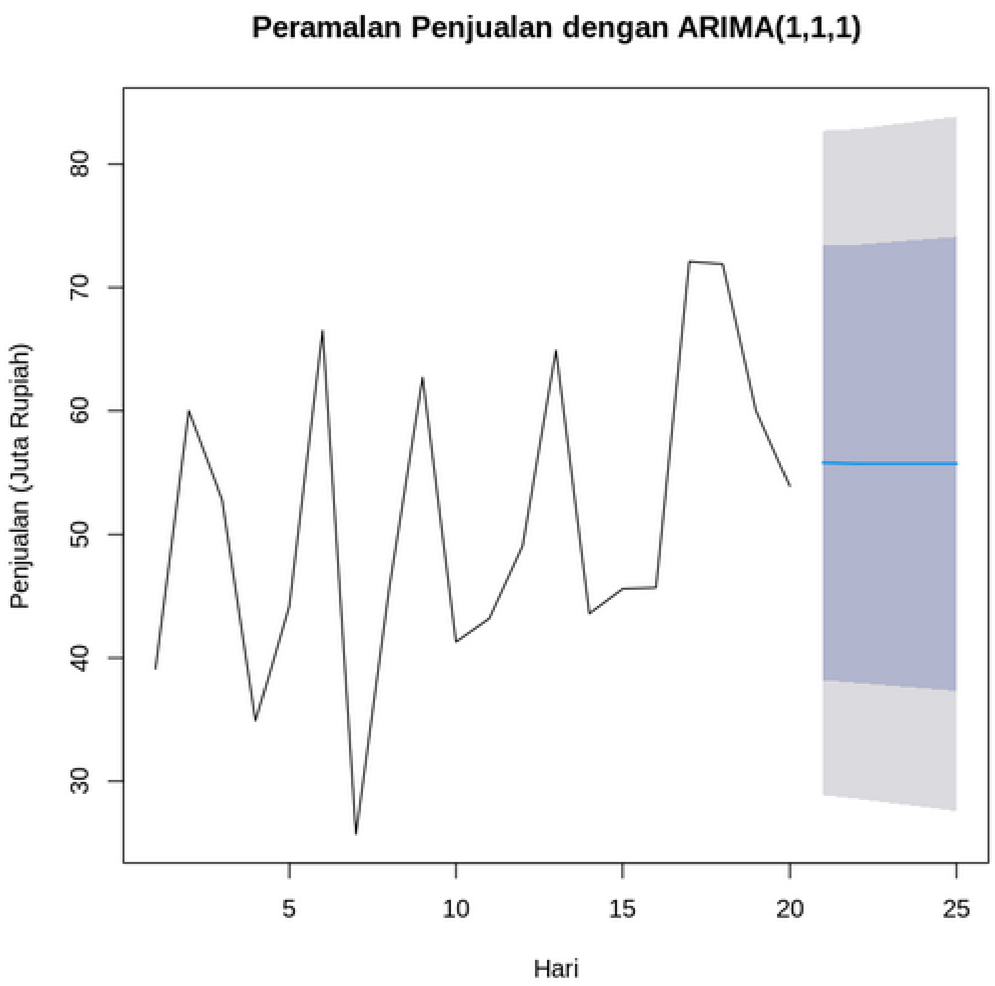
A. PEMBAHASAN LISTING

Praktik

1. Praktik 1

```r
#Praktik 1
#1 Persiapan Data

# Install dan Load package
install.packages("forecast")
install.packages("tseries")
library(forecast)
library(tseries)

# Input data
penjualan <- c(39.1, 60.0, 52.8, 34.9, 44.2, 66.5, 25.7, 45.7, 62.7, 41.3,
               43.2, 49.1, 64.9, 43.6, 45.6, 45.7, 72.1, 71.9, 60.0, 53.9)
penjualan_ts <- ts(penjualan, frequency = 1)
```

```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

also installing the dependencies 'xts', 'TTR', 'quadprog', 'quantmod', 'colorspace', 'fracdiff', 'lmtest', 'timeDate', 'tseries', 'urca', 'zoo', 'RcppArmadillo'


Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

Registered S3 method overwritten by 'quantmod':
  method            from
  as.zoo.data.frame zoo
```



```r
#5 Peramalan 5 hari kedepan

# Lakukan peramalan
forecast_result <- forecast(model, h = 5)

# Tampilkan hasil
print(forecast_result)

# Visualisasi
plot(forecast_result,
     main = "Peramalan Penjualan dengan ARIMA(1,1,1)",
     xlab = "Hari",
     ylab = "Penjualan (Juta Rupiah)")
```

```
   Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
21       55.79022 38.18952 73.39091 28.87228 82.70816
22       55.71433 37.97605 73.45261 28.58597 82.84269
23       55.71737 37.75566 73.67909 28.24730 83.18745
24       55.71725 37.53869 73.89582 27.91553 83.51897
25       55.71726 37.32425 74.11027 27.58758 83.84694
```

```r
#4 Pemodelan ARIMA (1,1,1)

# Fit model ARIMA(1,1,1)
model <- Arima(penjualan_ts, order = c(1,1,1))

# Tampilkan summary model
summary(model)
```

```
Series: penjualan_ts
ARIMA(1,1,1)

Coefficients:
          ar1      ma1
      -0.0401  -0.8345
s.e.   0.2642   0.1672

sigma^2 = 188.6:  log likelihood = -76.31
AIC=158.62   AICc=160.22   BIC=161.45

Training set error measures:
                   ME    RMSE     MAE        MPE     MAPE      MASE        ACF1
Training set 2.990164 12.6606 9.65858 -0.2355515 19.55572 0.6837296 -0.08214198
```

```r
#3 Transformasi Data (Karena Tidak Stasionary)

# Differencing orde 1
diff_penjualan <- diff(penjualan_ts)

# Cek stasioneritas setelah differencing
adf.test(diff_penjualan)  # p-value < 0.05 berarti stasioner
```

```
Warning message in adf.test(diff_penjualan):
“p-value smaller than printed p-value”

	Augmented Dickey-Fuller Test

data:  diff_penjualan
Dickey-Fuller = -5.1789, Lag order = 2, p-value = 0.01
alternative hypothesis: stationary
```

```r
#2 Cek Stasioneritas Data

# Uji Augmented Dickey-Fuller
adf.test(penjualan_ts)
```

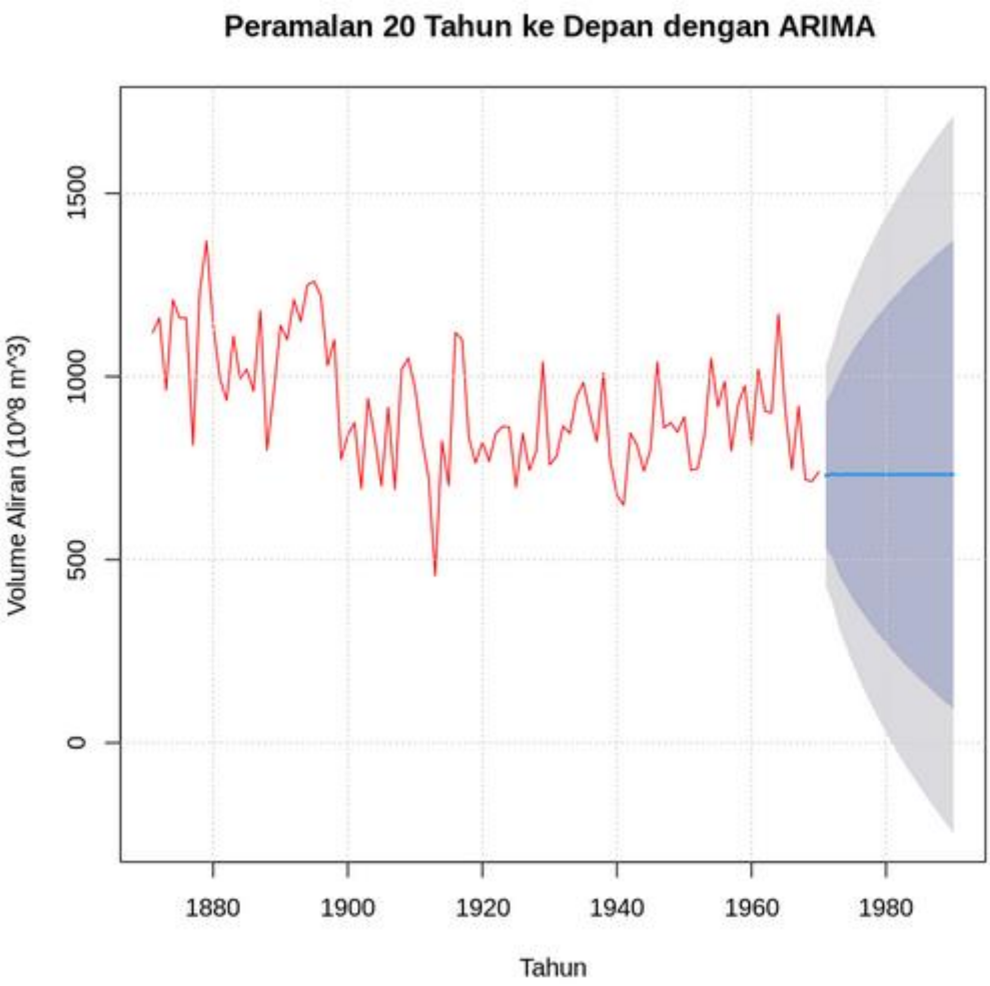
```
	Augmented Dickey-Fuller Test

data:  penjualan_ts
Dickey-Fuller = -3.1152, Lag order = 2, p-value = 0.1475
alternative hypothesis: stationary
```

Pembahasan : Pada praktik pertama, pemodelan dilakukan pada data penjualan selama 20 hari dengan model ARIMA(1,1,1) setelah konfirmasi ketidakstasioneran data melalui uji ADF (p-value 0,1475). Setelah differencing orde pertama, data menjadi stasioner (p-value < 0,01). Model yang dihasilkan menunjukkan nilai AIC 158,62, dengan peramalan 5 hari ke depan menghasilkan prediksi sekitar 55,7 juta rupiah per hari, disertai interval kepercayaan yang semakin melebar seiring horizon peramalan.

2. **Praktik 2**

```
#Praktik 2

#1 Load dataset bawaan
data(Nile)
ts_data <- Nile

# Tampilkan 10 observasi pertama
head(ts_data, 10)
```

A Time Series:
1120 · 1160 · 963 · 1210 · 1160 · 1160 · 813 · 1230 · 1370 · 1140

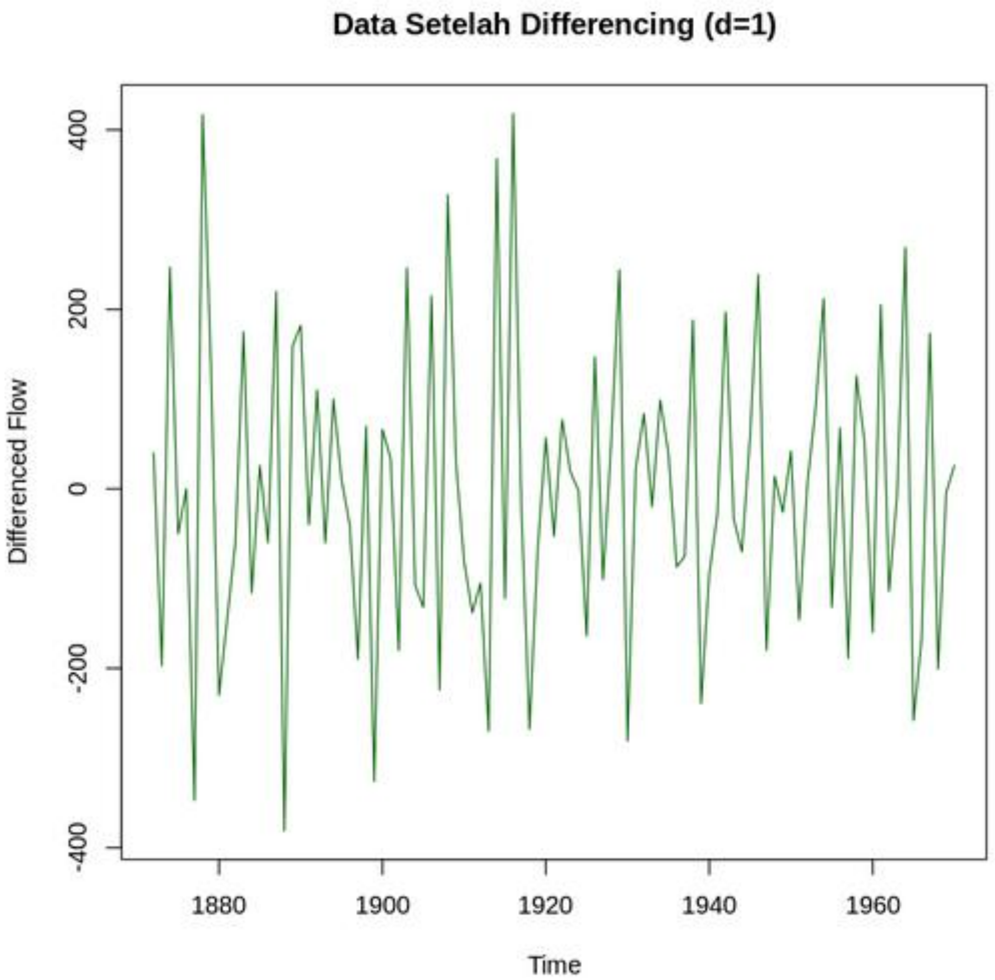


```
     Point Forecast       Lo 80     Hi 80       Lo 95    Hi 95
1971       729.6641 533.25793  926.0703  429.286768 1030.041
1972       733.7730 504.47641  963.0696  383.094135 1084.452
1973       732.1396 458.48404 1005.7951  313.619551 1150.660
1974       732.7889 426.74320 1038.8346  264.732394 1200.845
1975       732.5308 395.17496 1069.8866  216.589589 1248.472
1976       732.6334 367.38802 1097.8788  174.038796 1291.228
1977       732.5926 341.16310 1124.0221  133.952845 1331.232
1978       732.6088 316.74434 1148.4733   96.598990 1368.619
1979       732.6024 293.62185 1171.5829   61.239603 1403.965
1980       732.6049 271.68103 1193.5288   27.682659 1437.527
1981       732.6039 250.72921 1214.4786   -4.359848 1469.568
1982       732.6043 230.65466 1234.5540  -35.061429 1500.270
1983       732.6042 211.35130 1253.8570  -64.583289 1529.792
1984       732.6042 192.73824 1272.4702  -93.049551 1558.258
1985       732.6042 174.74565 1290.4627 -120.566828 1585.775
1986       732.6042 157.31560 1307.8928 -147.223796 1612.432
1987       732.6042 140.39830 1324.8101 -173.096566 1638.305
1988       732.6042 123.95104 1341.2574 -198.250473 1663.459
1989       732.6042 107.93668 1357.2717 -222.742323 1687.951
1990       732.6042  92.32274 1372.8857 -246.621785 1711.830
```

```
#6 Peramalan 20 tahun ke depan
forecast_20 <- forecast(model, h = 20)

# Plot hasil peramalan
plot(forecast_20,
     main = "Peramalan 20 Tahun ke Depan dengan ARIMA ",
     xlab = "Tahun",
     ylab = "Volume Aliran (10^8 m^3)",
     col = "red")
grid()

# Tampilkan nilai forecast
print(forecast_20)
# "Point Forecast" = prediksi utama
# Lo/Hi 80% & 95% = confidence interval
```

```r
#5 Fitting model ARIMA
install.packages("forecast")
library(forecast)

# model  ARIMA(1,1,0)
model <- arima(ts_data, order = c(1, 1, 0))
summary(model)
```

```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)


Call:
arima(x = ts_data, order = c(1, 1, 0))

Coefficients:
          ar1
      -0.3975
s.e.   0.0915

sigma^2 estimated as 23488:  log likelihood = -638.74,  aic = 1281.48

Training set error measures:
                    ME     RMSE     MAE       MPE     MAPE      MASE       ACF1
Training set -5.435748 152.4884 121.951 -2.514983 13.82784 0.9151874 -0.1020686
```



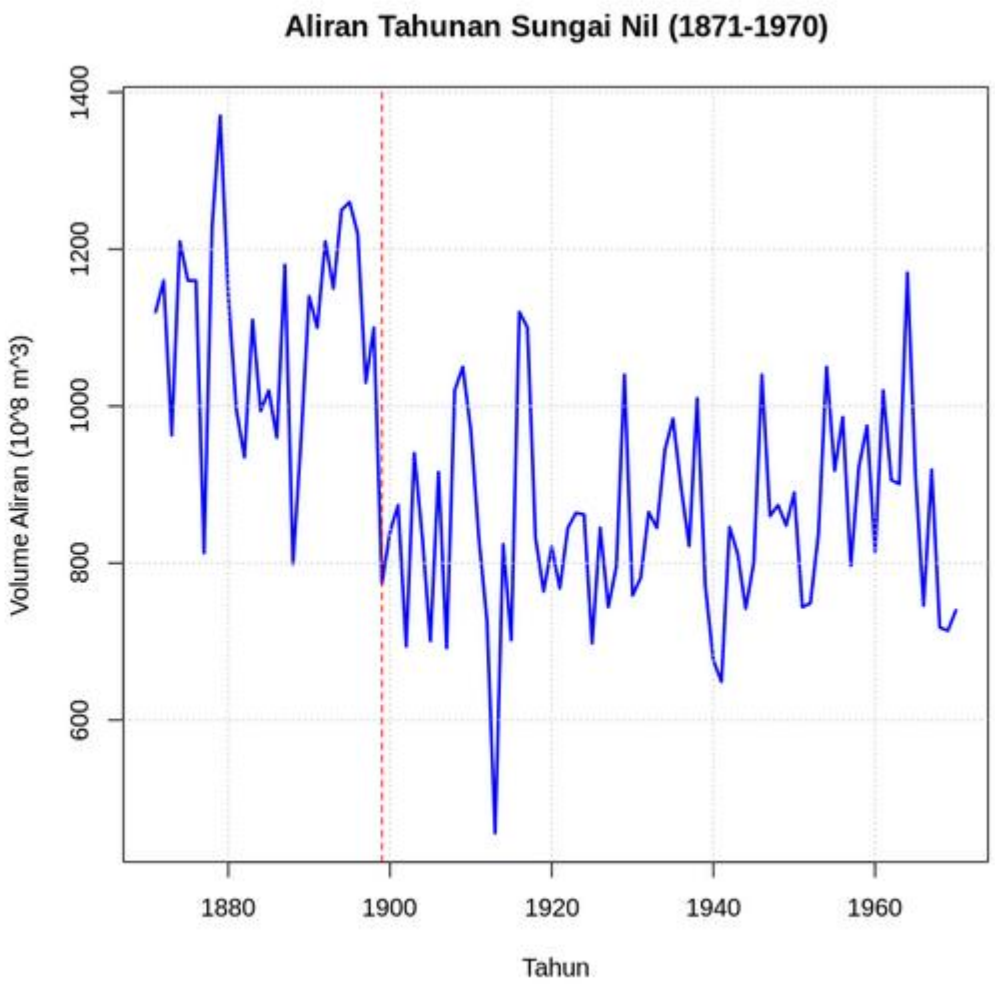
Data Setelah Differencing (d=1)
Differenced Flow
400
200
0
-200
-400
1880
1900
1920
1940
1960
Time

```r
#4 Differencing untuk membuat stasioner (d=1 )
ts_diff <- diff(ts_data)

# Plot data setelah differencing
plot(ts_diff,
     main = "Data Setelah Differencing (d=1)",
     ylab = "Differenced Flow",
     col = "darkgreen")

# Uji ADF lagi
adf.test(ts_diff)  # Harusnya p-value <0.05 → stasioner
```

```
Warning message in adf.test(ts_diff):
“p-value smaller than printed p-value”

    Augmented Dickey-Fuller Test

data:  ts_diff
Dickey-Fuller = -6.5924, Lag order = 4, p-value = 0.01
alternative hypothesis: stationary
```

```r
#3 Uji stasioneritas dengan ADF Test
install.packages("tseries")
library(tseries)

adf.test(ts_data)
# p-value biasanya >0.05 → tidak stasioner → perlu differencing
```

```
Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)


    Augmented Dickey-Fuller Test

data:  ts_data
Dickey-Fuller = -3.3657, Lag order = 4, p-value = 0.0642
alternative hypothesis: stationary
```



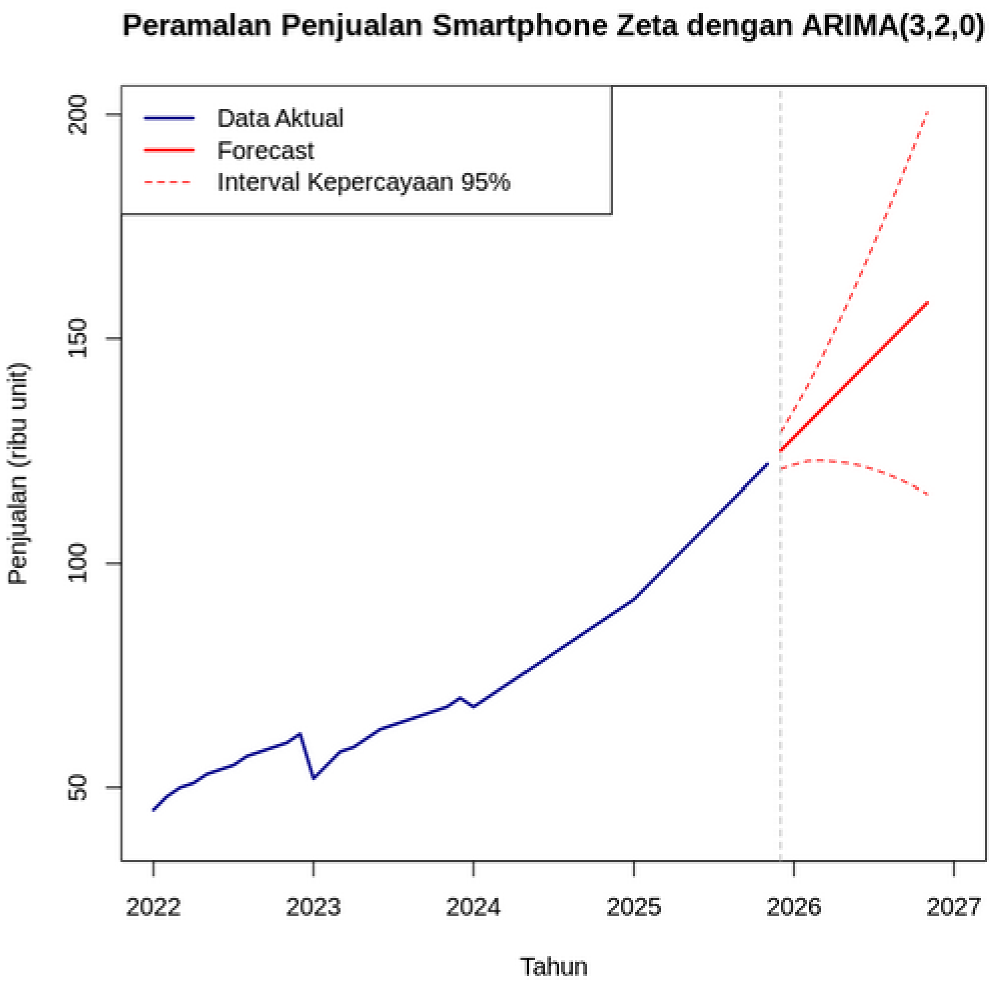

```
#2 Visualisasi data

plot(ts_data,
     main = "Aliran Tahunan Sungai Nil (1871-1970)",
     ylab = "Volume Aliran (10^8 m^3)",
     xlab = "Tahun",
     col = "blue",
     lwd = 2)
grid()
abline(v = 1899, col = "red", lty = 2)
```

Pembahasan : analisis diterapkan pada data tahunan aliran Sungai Nil (1871–1970). Data asli juga tidak stasioner (p-value ADF 0,0642), sehingga dilakukan differencing orde satu untuk mencapai stasioneritas. Model ARIMA(1,1,0) yang difitkan menghasilkan koefisien AR1 -0,3975 dengan AIC 1281,48. Peramalan 20 tahun ke depan (1971–1990) menunjukkan pola yang stabil di sekitar 732,6 satuan volume aliran, dengan interval kepercayaan (80% dan 95%) yang semakin luas seiring waktu, mencerminkan peningkatan ketidakpastian dalam peramalan jangka panjang.

Latihan

1. **Latihan 1**

```
#Latihan 1

# Data penjualan
data_2023 <- c(210, 195, 220, 205, 218, 230, 225, 240, 235, 228, 215, 245)
data_2024 <- c(228, 212, 240, 222, 235, 248, 242, 260, 255, 250, 232, 268)
data_2025 <- c(245, 230, 258, NA, NA, NA, NA, NA, NA, NA, NA, NA)

# Gabungkan data lengkap
penjualan <- c(data_2023, data_2024, data_2025[1:3])

# Konversi ke time series (mulai Jan 2023, frekuensi bulanan)
ts_data <- ts(penjualan, start = c(2023, 1), frequency = 12)

# Fitting model ARIMA(2,1,0)
model <- arima(ts_data, order = c(2, 1, 0))

# Forecast 12 bulan ke depan
forecast_result <- predict(model, n.ahead = 12)

# Tampilkan hasil
cat("Hasil Peramalan 12 Bulan ke Depan:\n")
cat("=================================\n")
forecast_values <- round(forecast_result$pred, 1)
se_values <- round(forecast_result$se, 1)
```

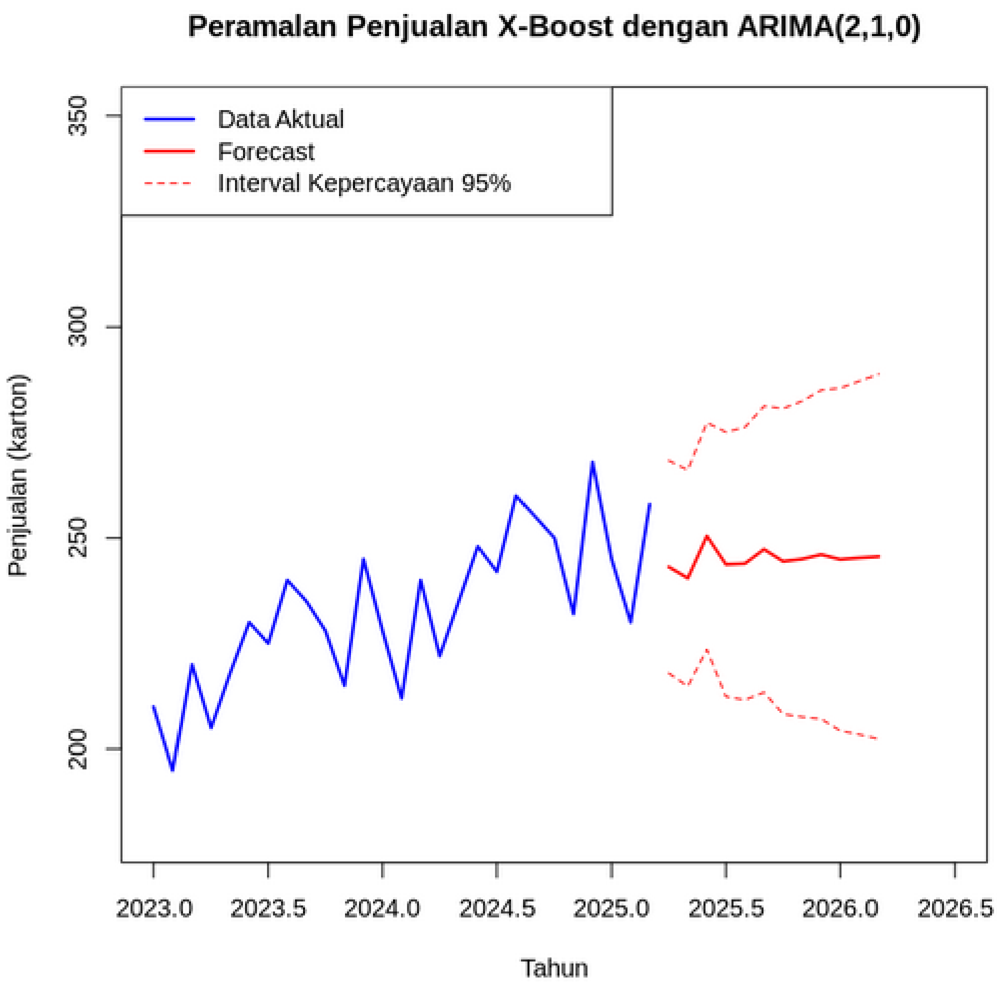


Peramalan Penjualan Smartphone Zeta dengan ARIMA(3,2,0)
Data Aktual
Forecast
Interval Kepercayaan 95%
Penjualan (ribu unit)
200
150
100
50
2022
2023
2024
2025
2026
2027
Tahun

```
Hasil Peramalan 12 Bulan ke Depan:
================================
Desember 2025: 125 ribu unit (±2.1)
Januari 2026: 128 ribu unit (±3.1)
Februari 2026: 131 ribu unit (±4.2)
Maret 2026: 134 ribu unit (±5.7)
April 2026: 137 ribu unit (±7.4)
Mei 2026: 140 ribu unit (±9)
Juni 2026: 143 ribu unit (±10.9)
Juli 2026: 146 ribu unit (±12.9)
Agustus 2026: 149 ribu unit (±14.9)
September 2026: 152 ribu unit (±17.1)
Oktober 2026: 155 ribu unit (±19.3)
November 2026: 158 ribu unit (±21.7)


Summary Model ARIMA(3,2,0):
============================

Call:
arima(x = ts_data, order = c(3, 2, 0))

Coefficients:
          ar1      ar2      ar3
      -0.8727  -0.6235  -0.2440
s.e.   0.1437   0.1674   0.1398

sigma^2 estimated as 4.273:  log likelihood = -97,  aic = 202.01
```

```r
bulan <- c("Desember 2025", "Januari 2026", "Februari 2026", "Maret 2026",
           "April 2026", "Mei 2026", "Juni 2026", "Juli 2026", "Agustus 2026",
           "September 2026", "Oktober 2026", "November 2026")

for(i in 1:12) {
  cat(bulan[i], ": ", forecast_values[i], " ribu unit (±", se_values[i], ")\n", sep = "")
}

# Plot data dan forecast
plot(ts_data, xlim = c(2022, 2027), ylim = c(40, 200),
     main = "Peramalan Penjualan Smartphone Zeta dengan ARIMA(3,2,0)",
     xlab = "Tahun", ylab = "Penjualan (ribu unit)", col = "darkblue", lwd = 2)
lines(forecast_result$pred, col = "red", lwd = 2)
lines(forecast_result$pred + 1.96*forecast_result$se, col = "red", lty = 2)
lines(forecast_result$pred - 1.96*forecast_result$se, col = "red", lty = 2)
abline(v = 2025 + 11/12, col = "gray", lty = 2) # Garis pemisah
legend("topleft", legend = c("Data Aktual", "Forecast", "Interval Kepercayaan 95%"),
       col = c("darkblue", "red", "red"), lty = c(1, 1, 2), lwd = c(2, 2, 1))

# Tampilkan summary model
cat("\n\nSummary Model ARIMA(3,2,0):\n")
cat("===========================\n")
print(model)
```

```r
#Latihan 2

# Data penjualan (dalam ribu unit)
data_2022 <- c(45, 48, 50, 51, 53, 54, 55, 57, 58, 59, 60, 62)
data_2023 <- c(52, 55, 58, 59, 61, 63, 64, 65, 66, 67, 68, 70)
data_2024 <- c(68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90)
data_2025 <- c(92, 95, 98, 101, 104, 107, 110, 113, 116, 119, 122, NA)

# Gabungkan data lengkap (sampai November 2025)
penjualan <- c(data_2022, data_2023, data_2024, data_2025[1:11])

# Konversi ke time series (mulai Jan 2022, frekuensi bulanan)
ts_data <- ts(penjualan, start = c(2022, 1), frequency = 12)

# Fitting model ARIMA(3,2,0)
model <- arima(ts_data, order = c(3, 2, 0))

# Forecast 12 bulan ke depan
forecast_result <- predict(model, n.ahead = 12)

# Tampilkan hasil
cat("Hasil Peramalan 12 Bulan ke Depan:\n")
cat("================================\n")
forecast_values <- round(forecast_result$pred, 1)
se_values <- round(forecast_result$se, 1)
```



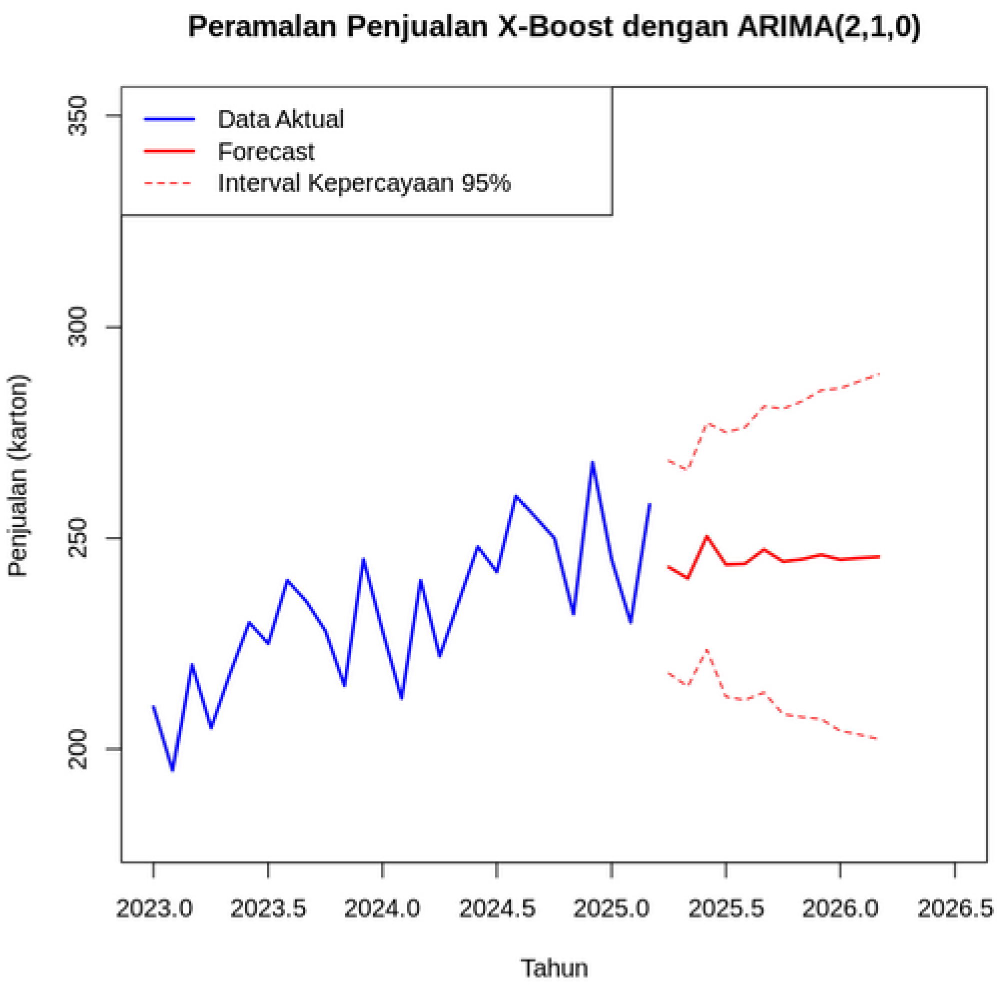

```
Hasil Peramalan 12 Bulan ke Depan:
================================
April 2025: 243.1 ± 12.9
Mei 2025: 240.5 ± 13.1
Juni 2025: 250.5 ± 13.7
Juli 2025: 243.7 ± 16
Agustus 2025: 243.9 ± 16.5
September 2025: 247.3 ± 17.3
Oktober 2025: 244.5 ± 18.5
November 2025: 245 ± 19.1
Desember 2025: 246.1 ± 19.9
Januari 2026: 244.9 ± 20.7
Februari 2026: 245.3 ± 21.4
Maret 2026: 245.6 ± 22.1
```

```r
bulan <- c("April 2025", "Mei 2025", "Juni 2025", "Juli 2025", "Agustus 2025",
           "September 2025", "Oktober 2025", "November 2025", "Desember 2025",
           "Januari 2026", "Februari 2026", "Maret 2026")

for(i in 1:12) {
  cat(bulan[i], ": ", forecast_values[i], " ± ", se_values[i], "\n", sep = "")
}

# Plot data dan forecast
plot(ts_data, xlim = c(2023, 2026.5), ylim = c(180, 350),
     main = "Peramalan Penjualan X-Boost dengan ARIMA(2,1,0)",
     xlab = "Tahun", ylab = "Penjualan (karton)", col = "blue", lwd = 2)
lines(forecast_result$pred, col = "red", lwd = 2)
lines(forecast_result$pred + 1.96*forecast_result$se, col = "red", lty = 2)
lines(forecast_result$pred - 1.96*forecast_result$se, col = "red", lty = 2)
legend("topleft", legend = c("Data Aktual", "Forecast", "Interval Kepercayaan 95%"),
       col = c("blue", "red", "red"), lty = c(1, 1, 2), lwd = c(2, 2, 1))
```

Pembahasan : Latihan 1, data penjualan tahun 2023–2025 (dengan data tidak lengkap untuk 2025) dimodelkan dengan ARIMA(2,1,0), namun output lengkap hasil peramalan 12 bulan tidak ditampilkan sepenuhnya dalam cuplikan. Pemilihan orde differencing (d=1) mengindikasikan adanya tren dalam data bulanan tersebut.

2. Latihan 2

```r
#Latihan 2

# Data penjualan (dalam ribu unit)
data_2022 <- c(45, 48, 50, 51, 53, 54, 55, 57, 58, 59, 60, 62)
data_2023 <- c(52, 55, 58, 59, 61, 63, 64, 65, 66, 67, 68, 70)
data_2024 <- c(68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90)
data_2025 <- c(92, 95, 98, 101, 104, 107, 110, 113, 116, 119, 122, NA)

# Gabungkan data lengkap (sampai November 2025)
penjualan <- c(data_2022, data_2023, data_2024, data_2025[1:11])

# Konversi ke time series (mulai Jan 2022, frekuensi bulanan)
ts_data <- ts(penjualan, start = c(2022, 1), frequency = 12)

# Fitting model ARIMA(3,2,0)
model <- arima(ts_data, order = c(3, 2, 0))

# Forecast 12 bulan ke depan
forecast_result <- predict(model, n.ahead = 12)

# Tampilkan hasil
cat("Hasil Peramalan 12 Bulan ke Depan:\n")
cat("================================\n")
forecast_values <- round(forecast_result$pred, 1)
se_values <- round(forecast_result$se, 1)
```




```
Hasil Peramalan 12 Bulan ke Depan:
================================
April 2025: 243.1 ± 12.9
Mei 2025: 240.5 ± 13.1
Juni 2025: 250.5 ± 13.7
Juli 2025: 243.7 ± 16
Agustus 2025: 243.9 ± 16.5
September 2025: 247.3 ± 17.3
Oktober 2025: 244.5 ± 18.5
November 2025: 245 ± 19.1
Desember 2025: 246.1 ± 19.9
Januari 2026: 244.9 ± 20.7
Februari 2026: 245.3 ± 21.4
Maret 2026: 245.6 ± 22.1
```

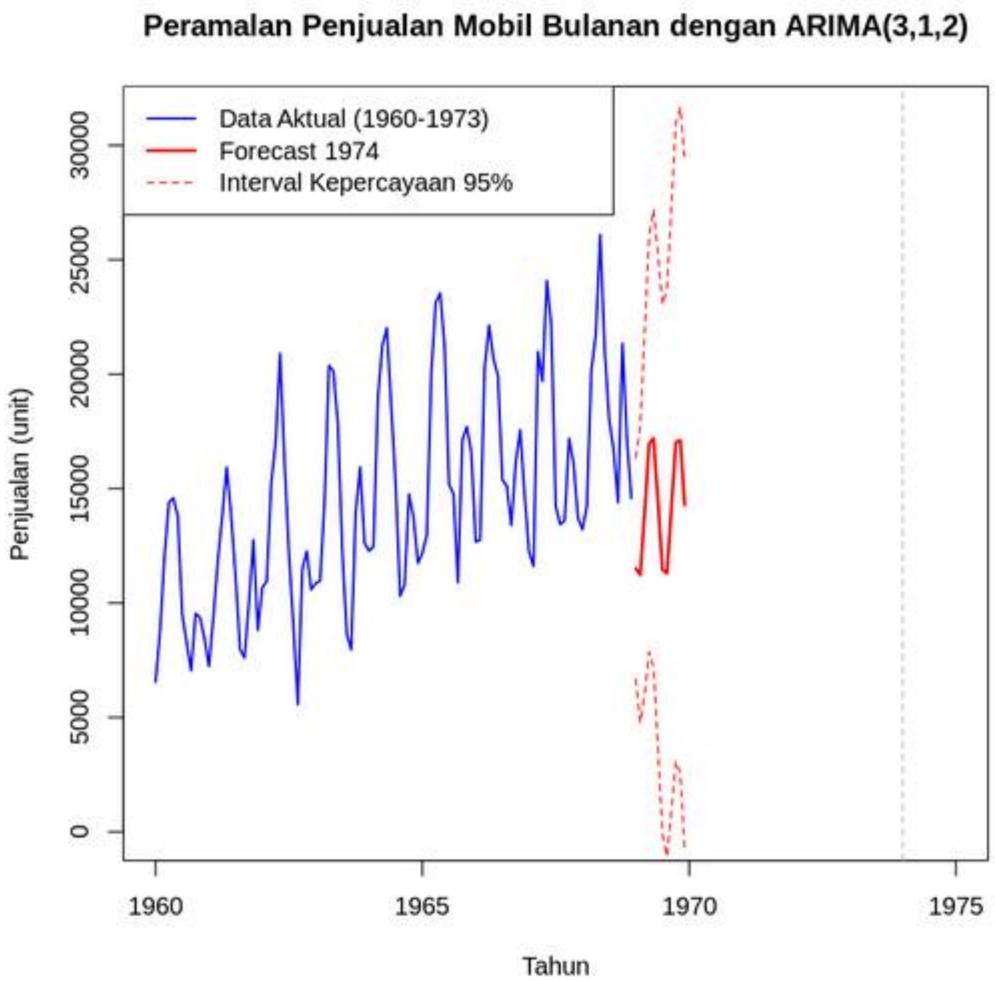
```r
bulan <- c("April 2025", "Mei 2025", "Juni 2025", "Juli 2025", "Agustus 2025",
           "September 2025", "Oktober 2025", "November 2025", "Desember 2025",
           "Januari 2026", "Februari 2026", "Maret 2026")

for(i in 1:12) {
  cat(bulan[i], ": ", forecast_values[i], " ± ", se_values[i], "\n", sep = "")
}

# Plot data dan forecast
plot(ts_data, xlim = c(2023, 2026.5), ylim = c(180, 350),
     main = "Peramalan Penjualan X-Boost dengan ARIMA(2,1,0)",
     xlab = "Tahun", ylab = "Penjualan (karton)", col = "blue", lwd = 2)
lines(forecast_result$pred, col = "red", lwd = 2)
lines(forecast_result$pred + 1.96*forecast_result$se, col = "red", lty = 2)
lines(forecast_result$pred - 1.96*forecast_result$se, col = "red", lty = 2)
legend("topleft", legend = c("Data Aktual", "Forecast", "Interval Kepercayaan 95%"),
       col = c("blue", "red", "red"), lty = c(1, 1, 2), lwd = c(2, 2, 1))
```

Pembahasan : data penjualan smartphone Zeta (2022–2025) menunjukkan pola pertumbuhan yang kuat, sehingga digunakan model **ARIMA(3,2,0)** dengan differencing orde dua (d=2) untuk menangani tren yang meningkat secara kuadratik. Hasil peramalan 12 bulan ke depan (Des 2025–Nov 2026) memproyeksikan kenaikan stabil dari 125 ribu unit menjadi 158 ribu unit, dengan interval kepercayaan yang melebar seiring waktu (dari ±2.1 hingga ±21.7), mencerminkan peningkatan ketidakpastian untuk periode yang lebih jauh. Model ini memiliki AIC 202.01, menunjukkan kecukupan fit yang baik.

B. PEMBAHASAN TUGAS

1. Tugas 1

```r
#Tugas 1

# Muat dataset langsung dari URL
url <- "https://raw.githubusercontent.com/jbrownlee/Datasets/master/monthly-car-sales.csv"
data <- read.csv(url)

# Konversi ke time series (mulai Jan 1960, frekuensi bulanan)
ts_data <- ts(data$Sales, start = c(1960, 1), frequency = 12)

# Fitting model ARIMA(3,1,2)
model <- arima(ts_data, order = c(3, 1, 2))

# Forecast 12 bulan ke depan
forecast_result <- predict(model, n.ahead = 12)

# Tampilkan hasil
cat("Hasil Peramalan 12 Bulan ke Depan (Jan 1974 - Des 1974):\n")
cat("========================================================\n")
forecast_values <- round(forecast_result$pred, 1)
se_values <- round(forecast_result$se, 1)

bulan <- c("Januari 1974", "Februari 1974", "Maret 1974", "April 1974",
           "Mei 1974", "Juni 1974", "Juli 1974", "Agustus 1974",
           "September 1974", "Oktober 1974", "November 1974", "Desember 1974")
```



```
Summary Model ARIMA(3,1,2):
============================

Call:
arima(x = ts_data, order = c(3, 1, 2))

Coefficients:
         ar1      ar2      ar3      ma1     ma2
      0.8205  -0.8342  -0.1660  -0.9382  0.9988
s.e.  0.1058   0.1011   0.1015   0.0814  0.1149

sigma^2 estimated as 5977148:  log likelihood = -990.43,  aic = 1992.85


Uji Normalitas Residual (Shapiro-Wilk):
=========================================

    Shapiro-Wilk normality test

data:  residuals(model)
W = 0.99398, p-value = 0.9214

Residual terdistribusi normal (p-value = 0.9214 )
```

```
Hasil Peramalan 12 Bulan ke Depan (Jan 1974 - Des 1974):
========================================================
Januari 1974: 11517.8 unit (±2465.9)
Februari 1974: 11222.5 unit (±3301.2)
Maret 1974: 13964.1 unit (±4044.5)
April 1974: 16967.6 unit (±4638.1)
Mei 1974: 17194.1 unit (±5108.8)
Juni 1974: 14419.5 unit (±5511.7)
Juli 1974: 11455.4 unit (±5907.8)
Agustus 1974: 11300.4 unit (±6324.2)
September 1974: 14106.2 unit (±6738.7)
Oktober 1974: 17029.6 unit (±7109.7)
November 1974: 17113.5 unit (±7423.9)
Desember 1974: 14278.1 unit (±7706.9)
```

```r
if(shapiro_test$p.value < 0.05) {
  cat("Residual tidak terdistribusi normal (p-value =", round(shapiro_test$p.value, 4), ")\n")
} else {
  cat("Residual terdistribusi normal (p-value =", round(shapiro_test$p.value, 4), ")\n")
}
```

```r
for(i in 1:12) {
  cat(bulan[i], ": ", forecast_values[i], " unit (±", se_values[i], ")\n", sep = "")
}

# Plot data dan forecast
plot(ts_data, xlim = c(1960, 1975), ylim = c(0, max(ts_data, forecast_result$pred)*1.2),
     main = "Peramalan Penjualan Mobil Bulanan dengan ARIMA(3,1,2)",
     xlab = "Tahun", ylab = "Penjualan (unit)", col = "blue", lwd = 1.5)
lines(forecast_result$pred, col = "red", lwd = 2)
lines(forecast_result$pred + 1.96*forecast_result$se, col = "red", lty = 2)
lines(forecast_result$pred - 1.96*forecast_result$se, col = "red", lty = 2)
abline(v = 1974, col = "gray", lty = 2) # Garis pemisah
legend("topleft", legend = c("Data Aktual (1960-1973)", "Forecast 1974", "Interval Kepercayaan 95%"),
       col = c("blue", "red", "red"), lty = c(1, 1, 2), lwd = c(1.5, 2, 1))

# Tampilkan summary model
cat("\n\nSummary Model ARIMA(3,1,2):\n")
cat("============================\n")
print(model)

# Analisis residual
cat("\n\nUji Normalitas Residual (Shapiro-Wilk):\n")
cat("========================================\n")
shapiro_test <- shapiro.test(residuals(model))
print(shapiro_test)
```

Pembahasan : Tugas 1, analisis dilakukan pada dataset penjualan mobil bulanan (1960–1973) dengan model ARIMA(3,1,2). Hasil peramalan untuk tahun 1974 menunjukkan fluktuasi musiman yang jelas, dengan puncak penjualan diprediksi terjadi pada April–Mei (sekitar 17.000 unit) dan September–Oktober. Uji normalitas residual mengonfirmasi bahwa residual terdistribusi normal (p-value Shapiro-Wilk = 0.9214), yang memenuhi salah satu asumsi penting dalam pemodelan ARIMA. Nilai AIC 1992.85 dan sigma^2 yang relatif besar (5.977.148) mencerminkan variabilitas data yang tinggi.

C. KESIMPULAN

pentingnya identifikasi orde differencing (d) yang sesuai untuk mencapai stasioneritas, terutama ketika data memiliki tren yang kuat atau musiman. Semakin panjang horizon peramalan, interval kepercayaan yang semakin melebar menjadi pengingat bahwa ketepatan prediksi akan menurun seiring waktu. Pemilihan orde AR dan MA yang berbeda (seperti ARIMA(3,2,0) vs ARIMA(3,1,2)) juga menunjukkan fleksibilitas model ARIMA dalam menangani karakteristik data yang beragam

D. LAMPIRAN

NIM : 245410034
Nama : Irvan Cahya Nugraha
Kelas : IF-1

LISTING PRAKTIKUM STATISTIKA
MODUL